# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME:** CSA04 **COURSE CODE:** Operating Systems

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

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**QUESTIONS: 1** **Total Marks:100**

Annexure A

**Concept Mapping**

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***Code of Conduct:***

*I* VISHAL BALAJI.J (192525014 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

95/100

Annexure A

## Concept Mapping

**Question 1**
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

**Question 2**
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

**Question 3**
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

**Question 4**
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

**Question 5**
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

19/10/26

Assessment Tool - 2

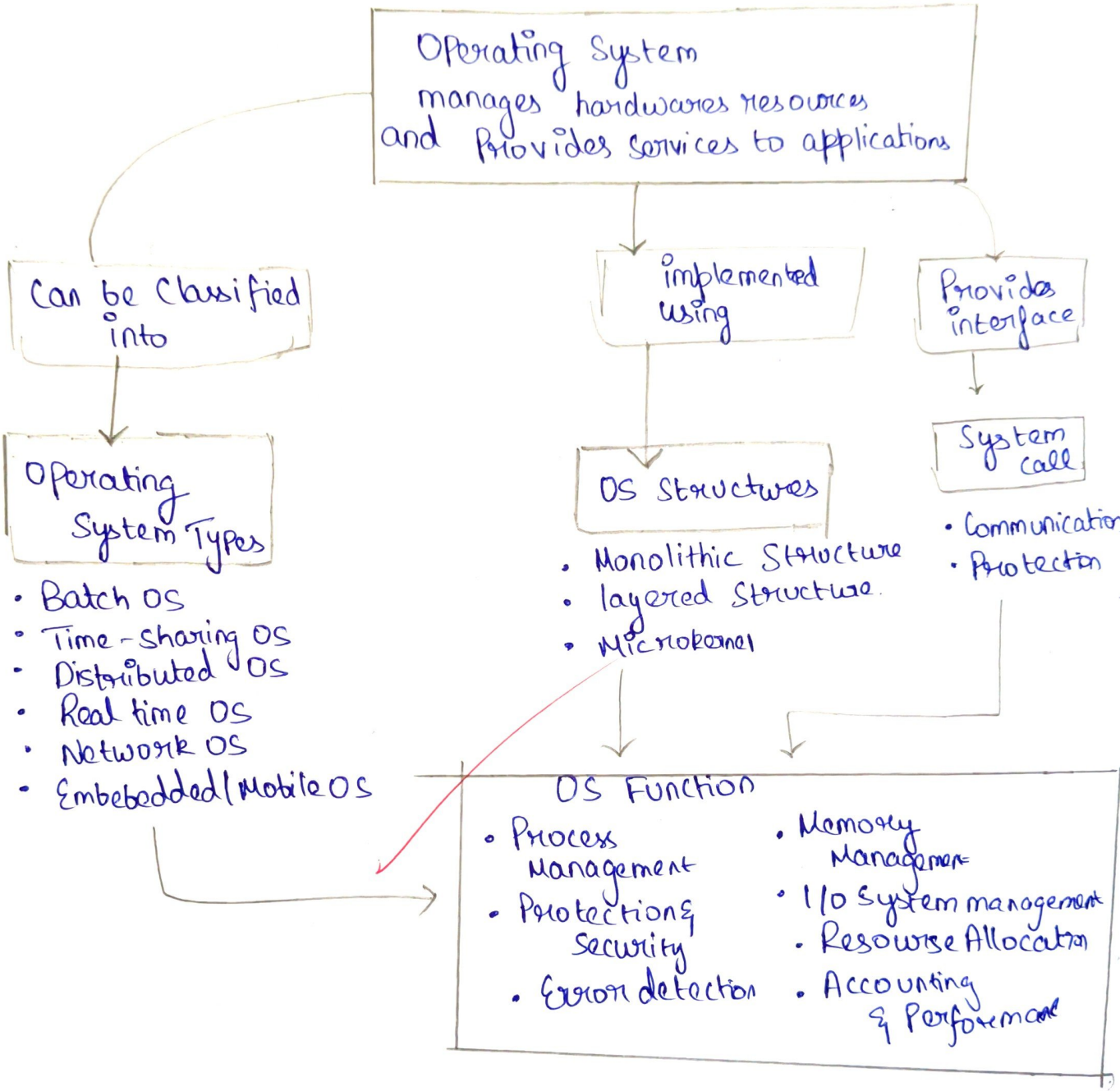
Operating System

CSA - 01

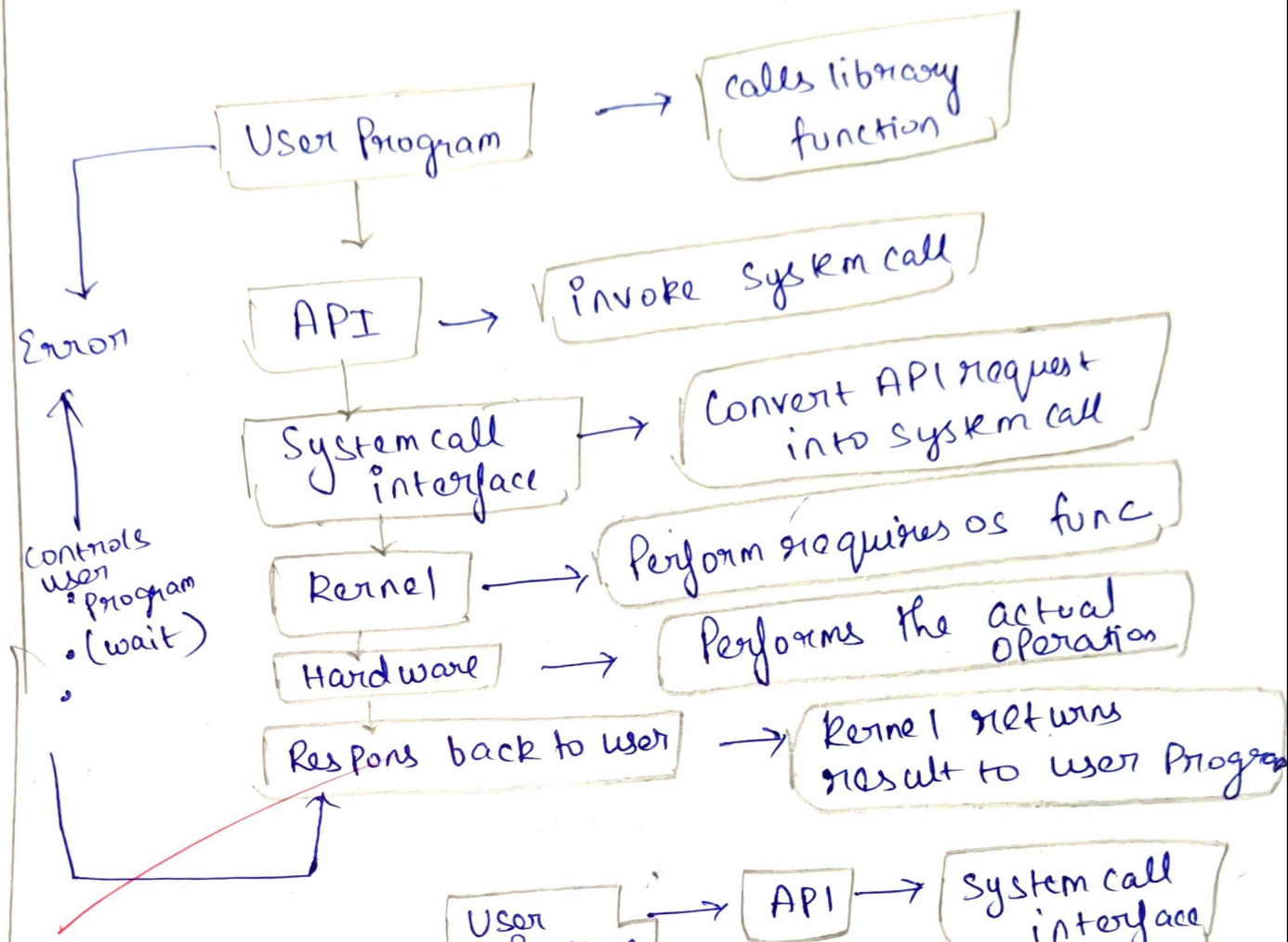
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Concept - Map

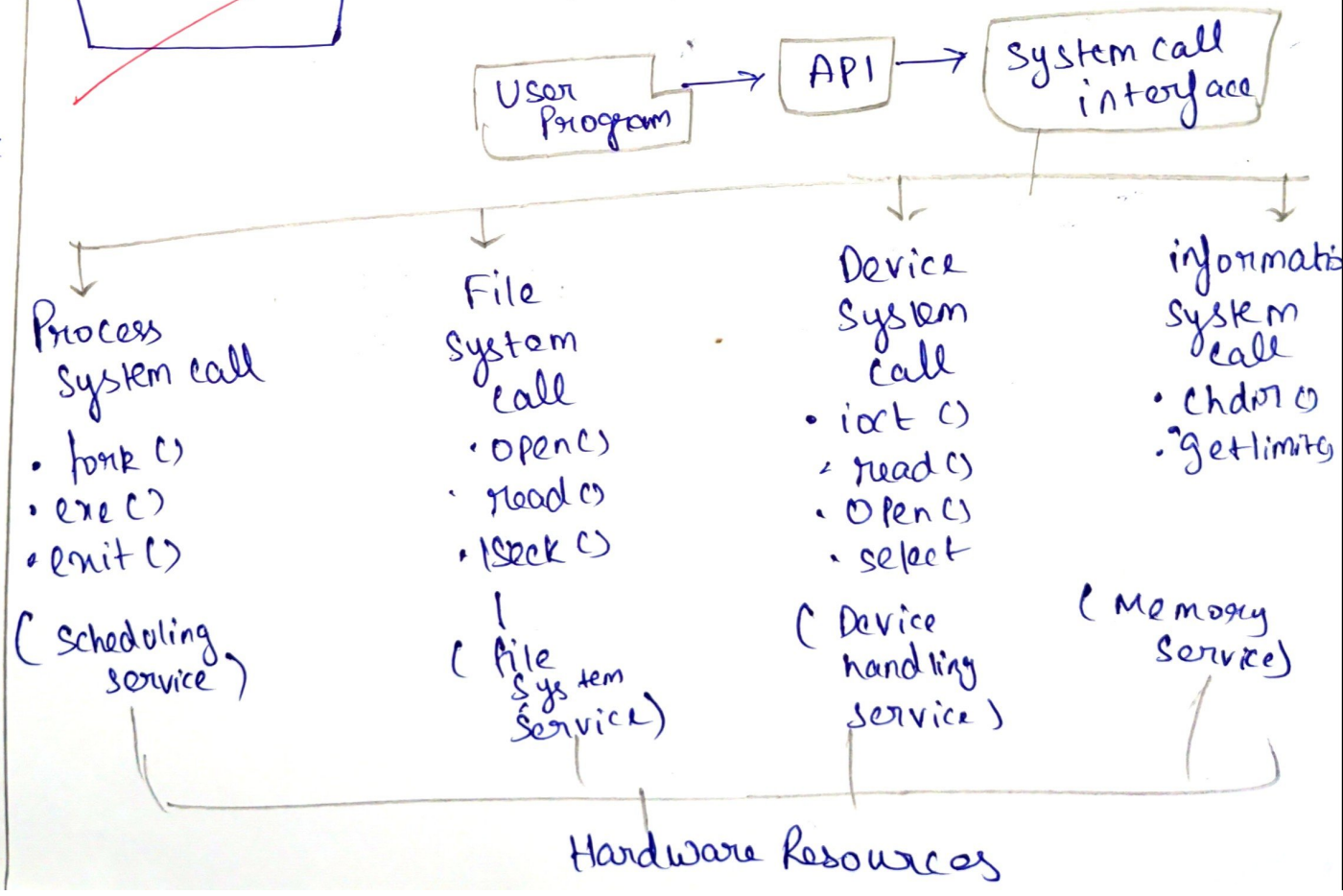
Q-1



Q-4 Execution flow (user Program to response




Q-5



Assessment Tool - 2

VISHAL BALAJI.J

## RUBRICS FOR CALCULATION

| Concept Mapping | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
| Concept Identification | 25% | Identifies all key concepts from literature accurately and comprehensively | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts |
| Linkages | 25% | Establishes clear, meaningful, and accurate relationships between concepts | Shows correct relationships with minor gaps | Relationships are weak, unclear, or partially incorrect | No meaningful relationships established |
| Organization | 20% | Concepts are well-structured hierarchically with logical flow and grouping | Mostly organized with minor structural issues | Some organization but lacks clear hierarchy | No logical organization or structure |
| Integration of Literature | 20% | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature | Limited integration; mostly isolated concepts | No integration of literature evidence |
| Clarity | 10% | Concept map is clear, neat, visually structured, and easy to interpret | Generally clear with minor readability issues | Clarity is reduced; difficult to interpret in parts | Poorly presented and hard to understand |